Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
10.	*(a)*			Total percentage of ^{29}Si and ^{30}Si is 100 – 92.2 = 7.8 % of $^{29}Si = \frac{2 \times 7.8}{3} = 5.2$ and % of $^{30}Si = \frac{1 \times 7.8}{3} = 2.6$ (1) $\therefore A_r = \frac{(28 \times 92.2) + (29 \times 5.2) + (30 \times 2.6)}{100}$ (1) $\therefore A_r = \frac{2582 + 151 + 78}{100} = \frac{2811}{100} = 28.1$ (1) Answer only – no mark ecf possible		2	1	3	1 1	
	(b)	(i)		Tetrahedral	1			1		
		(ii)		There are no free electrons or ions to carry the charge	1			1		
		(iii)		Any of the following • There are no electronegativity differences in the Si—Si bond • All the bonding electrons are shared equally between the four Si atoms • Si cannot lose or gain 4 electrons			1	1		

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					AO1	AO2	AO3	Total	Maths	Prac
10.	*(c)*			M_r HF is 20.01 Solution contains 500g HF in 1000g solution (using V = m/D) 1000 g of the solution has a volume of 855 cm^3 or Number of moles of HF in 1000g / 855 cm^3 solution is $\frac{500}{20.01} = 24.98$ (1) 855 cm^3 contain 24.98 mol $\therefore$ Concentration of HF = 29.2 mol dm^{-3} (1) ecf possible		2		2	2	
	(d)	(i)		(There are 6 bonding pairs of electrons and no lone pairs –) position of minimum repulsion taken up (1) Drawing shows clear octahedral shape (1) Bond angle is 90^o equatorial / equatorial or 90^o equatorial / vertical (accept 180^o if vertical bonds only considered) (1)	1	2		3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.		(ii)		1 mol of H_2SiF_6 (144g) gives 6 mol of F^- ions (6 × 19g) = 114g (1) ∴ 114 mg of fluoride ions from 144 mg H_2SiF_6 The increase in fluoride ion concentration needed is 0.76 – 0.15 = 0.61 mg dm^{-3} (1) Amount of H_2SiF_6 needed is $\frac{144 \times 0.61}{114}$ = 0.77 mg (1) Accept alternative method Award (3) for cao			3	3	2	
	(e)			$MgSiF_6$ M_r 166 ∴ Moles of $MgSiF_6$ = 2.6/166 = 1.566×10^{-2} (1) $[H^+]$ is ∴ 4 × 1.566×10^{-2} = 0.06265 mol dm^{-3} (1) pH = $-\log_{10}[H^+]$ = 1.20 (1) ecf possible		3		3	2	
				Question 10 total	**3**	**9**	**5**	**17**	**8**	**0**